

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0

- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Java Spring Boot, Neo4j, and PostgreSQL stack.
- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Authored an AWS deployment script and managed the environment as code with Terraform, replacing a manual 30-minute multi-service bring-up with a single-command, version-controlled provision.

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Ran the node graph byte-identical in Gazebo and on a physical RoboMaster EP Core, driving the real robot over a custom C++ WiFi driver at a 30 Hz control loop with sub-20ms command latency.
- Built the verifier as a C++ ROS2 node running per-step geometric and semantic checks on each motion plan, integrating nav2's MPPI controller rather than hand-rolling one to keep the C++ work on security, not motion control.
- Reproduced a published adversarial-robotics attack loop (attacker, target, judge, syntax-checker) as the baseline, then drove the decomposition attack to an 80% success rate against the per-step triage gate.

Robostangs (FRC Team 548)

Software Captain

Jun 2023 – May 2024

Northville, MI

- Hardened pose estimation against bad vision data by gating camera measurements on tag count, target area, and angular velocity before fusing them, rejecting estimates that would otherwise poison the localization filter.
- Led the software and electrical teams (5 and 6 engineers) for a competition robot that reached the FIRST World Championship and placed top 16 in its division in Houston, the program's strongest world-stage finish since 2014.

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Shipped the platform continuously from main on a GitHub Actions pipeline that applies database migrations and deploys edge functions on every push, taking the team from manual changes to reviewed continuous deploys.
- Built event check-in as a single atomic database RPC that validates a QR token, records attendance, and awards points server-side, making door-time check-ins concurrency-safe and idempotent, sustaining 150+ check-ins per event.

PROJECTS

Dadei | *Python, FastAPI, React, Redis, Docker*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Built a React client that streams microphone audio up over a WebSocket with client-side Web Audio processing, receiving incremental live captions back in under 300ms as the backend transcribes each utterance.
- Routed all model-proposed side effects through one approval chokepoint with a cancelable countdown, giving every irreversible action one auditable seam between the LLM and the outside world.

TECHNICAL SKILLS

Languages Python, Java, C++, JavaScript/TypeScript, SQL, Bash

Frameworks React, Spring Boot, FastAPI

Databases PostgreSQL, Neo4j, Redis

Cloud & Infrastructure AWS, Docker, Terraform, GitHub Actions, Linux

Tools Git, Linux, Postman